

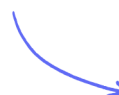
Multiple Choice Questions

Covalent Bonding & Structure

Evidence of Covalent Bonding / Covalent Bonding / Carbon Allotropes /
Electronegativity / VSEPR Theory / Predicting Shapes & Angles

Easy (2 questions)	/2
Medium (5 questions)	/5
Total Marks	/7

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Easy Questions

1 Which substance has a giant lattice of atoms?

- A. diamond
- B. ice
- C. poly(ethene)
- D. sodium chloride

(1 mark)

2 Which molecule is planar?

- A. CF_4
- B. C_2F_4
- C. PF_5
- D. SF_6

(1 mark)

Medium Questions

- 1 The hydrogen ion, H^+ , bonds to a water molecule forming an H_3O^+ ion.

For H_3O^+ , what shape and H-O-H bond angle are predicted by the electron-pair repulsion theory?

- A. trigonal planar 120°
- B. trigonal planar 117.5°
- C. trigonal pyramidal 107°
- D. trigonal pyramidal 104.5°

(1 mark)

- 2 Which of these molecules is **not** polar?

- A. CO_2
- B. HCl
- C. H_2O
- D. NH_3

(1 mark)

- 3 Which row gives the correct polarities of the S—F bond and the SF_6 molecule?

(1)

		Polarity of S—F bond	Polarity of SF ₆ molecule
☐	A	polar	polar
☐	B	polar	non-polar
☐	C	non-polar	polar
☐	D	non-polar	non-polar

(1 mark)

4 Which compound has bonds that are the most polar?

- A.** H₂O
- B.** H₂S
- C.** NH₃
- D.** PH₃

(1 mark)

5 Which of these molecules is the **most** polar?

- A.** HF
- B.** OF₂
- C.** BF₃
- D.** CF₄

(1 mark)